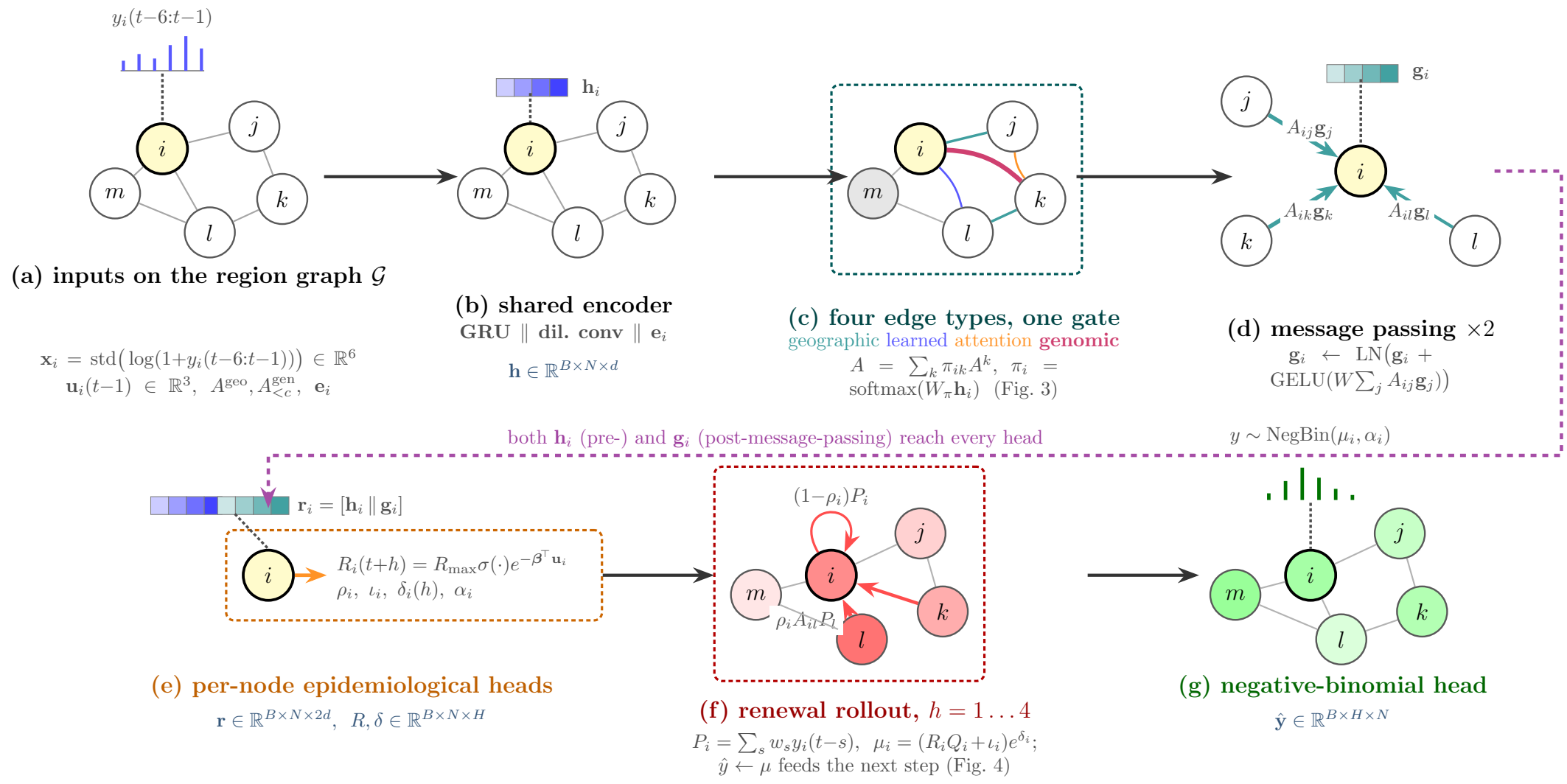


(a) inputs on the region graph $\mathcal{G}$

$\mathbf{x}_i = \mathrm{std}\big(\log(1{+}y_i(t{-}6{:}t{-}1))\big) \in \mathbb{R}^6$
$\mathbf{u}_i(t{-}1) \in \mathbb{R}^3,\ A^{\mathrm{geo}}, A^{\mathrm{gen}}_{<c},\ \mathbf{e}_i$

(b) shared encoder

GRU $\|$ dil. conv $\|$ $\mathbf{e}_i$

$\mathbf{h} \in \mathbb{R}^{B\times N\times d}$

(c) four edge types, one gate

geographic learned attention genomic

$A = \sum_k \pi_{ik} A^k,\ \pi_i = \mathrm{softmax}(W_\pi \mathbf{h}_i)$ (Fig. 3)

(d) message passing $\times 2$

$\mathbf{g}_i \leftarrow \mathrm{LN}\big(\mathbf{g}_i + \mathrm{GELU}(W \sum_j A_{ij}\mathbf{g}_j)\big)$

$y \sim \mathrm{NegBin}(\mu_i, \alpha_i)$

(e) per-node epidemiological heads

$\mathbf{r} \in \mathbb{R}^{B\times N\times 2d},\ R, \delta \in \mathbb{R}^{B\times N\times H}$

(f) renewal rollout, $h = 1 \ldots 4$

$P_i = \sum_s w_s y_i(t{-}s),\ \mu_i = (R_i Q_i + \iota_i)e^{\delta_i}$;
$\hat{y} \leftarrow \mu$ feeds the next step (Fig. 4)

(g) negative-binomial head

$\hat{\mathbf{y}} \in \mathbb{R}^{B\times H\times N}$